

Yanghonghui Chen

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EDUCATION

University of California, Los Angeles

Los Angeles US

Master of Science in Electrical and Computer Engineering – GPA: 3.9/4.0

Expected Sept. 2025 – Dec. 2026

University of Illinois Urbana-Champaign

Urbana US

Bachelor of Science in Electrical Engineering - GPA: 3.82/4.0

Sept. 2021 – Jun. 2025

Zhejiang University

Haining China

Bachelor of Engineering in Electrical Engineering - GPA: 3.94/4.0

Sept.2021 – Jun. 2025

PROJECT EXPERIENCES

Neural Speech Decoding System (Brain-to-Text)

Course Project (Advisor: Prof. Jonathan Kao)

Oct. 2025 - Dec. 2025

Keywords: *PyTorch, LSTM/GRU, BCI, Real-time Decoding, CTC Loss, Google Cloud Platform (GCP)*

- Engineered a real-time, uni-directional LSTM decoder with a specialized non-linear post-stack (Linear-LayerNorm-GELU) to capture long-range temporal dependencies. Enforced strict causal constraints to ensure viability for live Brain-Computer Interface applications, reducing error rates from 23.6% to 19.59%.
- Developed a robust training pipeline using Focal CTC Loss to address severe phoneme class imbalance in intracranial neural data. Stabilized convergence on Google Cloud (Tesla T4) through Gradient Clipping, AdamW optimization, and a sequential learning rate scheduler (Warmup + Cosine Annealing).

High-Speed MASH-1-1-1 Delta-Sigma Modulator Design in 16nm FinFET

Course Project (Advisor: Prof. Hooman Darabi)

Oct. 2025 - Dec. 2025

Keywords: *Verilog, ASIC Design Flow, Logic Synthesis, Static Timing Analysis (STA), Synopsys DC/PrimeTime, TCL, Low Power*

- Engineered a high-speed, pipelined MASH-1-1-1 digital modulator targeting 500 MHz clock frequency. Implemented a robust noise-shaping logic using 5-bit signed arithmetic and bit-extension techniques to handle dynamic ranges (-3 to +4) without overflow, ensuring unconditional stability.
- Synthesized the design using TSMC 16nm FinFET technology with Synopsys Design Compiler. Achieved timing closure across Multi-Mode Multi-Corner (MMMC) conditions (Setup @ SS/125°C, Hold @ FF/-40°C) by developing custom TCL scripts to automatically resolve critical hold-time violations via buffer insertion.
- Conducted gate-level simulations in ModelSim to generate switching activity (VCD) files. Performed high-precision sign-off power analysis using Synopsys PrimeTime PX, achieving a finalized power consumption of 0.178 mW and validating the design against pre-synthesis estimates.

Four-Axis Vacuum Stage for Advanced Nano-Manufacturing

Senior Design (Advisor: Prof. Oleskiy Penkov)

Feb. 2025 - May 2025

Keywords: *STM32, RS-485, Stepper Motor Control, PCB Design, HMI, FSM, C/C++*

- Control System Design: Engineered an STM32-based control system for a 4-DOF robotic arm, using the RS-485 protocol for robust communication with four stepper motor drivers. Developed control logic for precise multi-axis motion and implemented a Finite State Machine (FSM) to automate complex coating sequences.
- Custom PCB Development: Independently designed and validated a custom PCB to integrate the power distribution (24V) and RS-485 communication bus. The design minimized signal interference and voltage drop, ensuring system reliability in a high-vacuum environment.
- HMI & User Interface: Created an intuitive HMI using a TFT touchscreen and physical buttons. The interface provides real-time monitoring of motor status (speed, position) and allows for one-touch execution of preset programs and emergency stops.

Scalable EEG Signal Classification via Federated Learning

Senior Thesis (Advisor: Prof. Howard Yang)

Feb. 2025 - May 2025

Keywords: *TensorFlow, Federated Learning, Personalized FL (pFedMe), EEG, Seizure Prediction*

- Designed and implemented a privacy-preserving pipeline for seizure prediction, processing EEG signals into Pearson Correlation Coefficient (PCC) matrices and training a CNN classifier using four distinct paradigms: Local, Centralized, FedAvg, and pFedMe.
- Demonstrated that standard FedAvg underperforms due to client data heterogeneity and severe class imbalance, while personalized pFedMe significantly improves average sensitivity and F1-scores over FedAvg by adapting to unique patient data.

- Conducted a detailed hyperparameter analysis of the pFedMe personalization weight (λ), revealing that smaller λ values are critical for minority class detection (sensitivity), whereas larger values cause performance to collapse by forcing adherence to a globally biased model.

Optimized Convolutional Layer Implementation Using CUDA

Course Project (Advisor: Prof. Volodymyr Kindratenko)

Oct. 2024 - Dec. 2024

Keywords: *CUDA, Parallel programming, GPU acceleration, Convolutional neural networks (CNNs)*

- Designed and implemented the forward pass of convolutional layers for a modified LeNet-5 architecture using CUDA, optimizing performance for deep learning tasks such as image classification and object detection.
- Implemented a GPU-based forward convolution with a structured Prolog-Kernel-Epilog approach, ensuring memory management, convolution computation, and output transfer, while matching CPU implementation correctness and optimizing performance using Nsight profiling tools.
- Applied advanced GPU programming techniques to optimize the implementation, including streams, GEMM kernels, and kernel fusion, to achieve a target inference time of $\leq 80\text{ms}$ for 10,000 images from the Fashion MNIST dataset.

RRAM-based Heterogeneous Processing for Multimodal Brain-Computer Interfaces

The University of Hong Kong (Advisor: Dr. Zhengwu Liu)

May 2024 - Jul. 2024

Keywords: *RRAM, Multimodal brain-computer interface, EEG, Compute-in-memory, FDT*

- Simulated representative heterogeneous processing paradigm of P300 signal recognition in Python by using resistive random-access memory (RRAM) with and without fixed parameter disturbance training (FDT).
- Contributed to combining the BCIC IV IIa and the P300 RSVP datasets and designing an RRAM-based multimodal recognizer that integrates components of the pre-trained EEGNet, CSP (Common Spatial Pattern), and a modality-fused classifier to create the multimodal settings.
- Achieved 2.83% higher accuracy using multimodal BCI with FDT than that without FDT and significantly outperformed the MI-alone and P300-alone results by 8.19% and 13.20%, respectively.

Raspberry Pi Based IoT System as a Private Chatbot

Course Project (Advisor: Prof. Deming Chen)

Mar. 2024 - May 2024

Keywords: *IoT System, Raspberry Pi, Machine Learning, Deep Learning*

- Developed an IoT system using Raspberry Pi 4 as a private chatbot with face detection and speaker recognition to guarantee privacy and personal conversations as well as interactions.
- Implemented MTCNN with ResNet and dlib-based face recognition, achieving better performance with the latter; trained the system with one hundred face images for live recognition.
- Built a custom residual neural network with Keras for speaker recognition, achieving 96% accuracy.
- Integrated a server-client architecture using Google Cloud for accelerated processing and implemented speech recognition and TTS for user interaction.

Multiplayer Action Game on FPGA: Crazy Arcade

Course Project (Advisor: Prof. Zuofu Cheng)

Mar. 2024 - May 2024

Keywords: *SystemVerilog, FPGAs, System-on-a-chip, MicroBlaze CPU, VGA*

- Used FPGA for real-time operations, integrating MicroBlaze CPU for game logic and keyboard input processing.
- Developed various modules in SystemVerilog to manage player movements, bomb mechanics, life counts, and game states, interacting through a system bus.
- Incorporated background music by PWM for sound generation, featuring distinct tracks for different game stages.

Cheat-Machine for Game 2048

Course Project (Advisor: Prof. Thomas Moon)

Mar. 2024 - May 2024

Keywords: *Embedded DSP, Real-time Signal Processing, Image Processing, Android Studio*

- Developed an app in Android Studio to analyze a live game of 2048, recognizing board digits using image processing.
- Employed efficient template matching for multi-digit recognition, using grayscale conversion, Canny edge detection, and perspective transformation to preprocess images. Used python packages to evaluate the workflow of the application.
- Built an AI engine with an Expecti-max Search algorithm to recommend the optimal move, focusing on corner placement strategies.
- Achieved high accuracy in digit recognition (100% when properly aligned) and consistent AI performance, reaching 1024 tile in 75% of simulations.

Hook&Hair Structure 3D-Printing based on Path Control and 4D Printing Experiment

Exploration

Zhejiang University (Advisor: Prof. Guanyun Wang)

Jun. 2023 - Aug. 2023

Keywords: 3D printing, Grasshopper, Rhino, FDM, Path-planning, 4D printing

- Developed 3D printing techniques for complex hook and hair structures using Rhino and Grasshopper for path planning, generating G-codes for customized printing paths instead of traditional FDM (Fused Deposition Modeling) methods.
- Accomplished applications including hooked ball-mitten toys and hairy objects, requiring precise path control to avoid defects and achieve intricate designs.
- Conducted experiments in 4D printing, modeling deformable planar objects in Fusion360 that transform into stereoscopic shapes when heated.

Over-the-Air-Computation Based Federated Learning Model Establishment & Simulation

Zhejiang University (Advisor: Prof. Howard Yang)

Apr. 2022 – Apr. 2023

Keywords: Edge Computing, Federated Learning, OFDM, Over-the-air Computing, Simulink, Machine Learning, Neural Networks

- Explored an innovative approach to utilizing private data from distributed databases to train shared models, ensuring user privacy while making use of the data.
- Set up an over-the-air-computation-based communication model in Simulink which could transmit and receive massive data gradients between federated users effectively.
- Combined Machine Learning models like Linear Regression and Deep Learning models like neural networks in MATLAB codes with communication models in Simulink to implement effective edge-computing models.
- Improved the model to adapt to the Large-scale applications by exploiting and modifying existing OFDM Communication Systems.

SKILLS

- **Programming & Core Languages:** Python, C/C++, SystemVerilog, Verilog, CUDA, MATLAB, SQL, TCL (for EDA scripting).
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras, Federated Learning (pFedMe), Deep Learning (CNN, LSTM/GRU), BCI Signal Processing, Google Cloud Platform (GCP).
- **Digital IC & FPGA Design:** ASIC Design Flow, Logic Synthesis (Synopsys DC), Static Timing Analysis (PrimeTime), Low Power Design, FPGA Prototyping (Xilinx/Vivado), ModelSim.
- **Embedded Systems & Hardware:** STM32 MCU, PCB Design, IoT System Integration, Raspberry Pi, Communication Protocols (RS-485, UART), 3D Printing & Path Planning.